

PRECIOUS AFRAM

GIS Analyst & Remote Sensing Specialist

+233 541 562 319 | aframprecious3@gmail.com | Sunyani, Ghana | precious-afram.github.io/portfolio

PROFESSIONAL SUMMARY

GIS Analyst and Remote Sensing Specialist with hands-on experience in land use/land cover change analysis, satellite imagery processing, and environmental monitoring. Proficient in Google Earth Engine (GEE), QGIS, ArcMap, and GPS-based field surveys. Led projects spanning a 20-year deforestation study in Berekum Municipality and a national-scale air quality analysis across all 16 regions of Ghana. Proven ability to translate complex spatial data into policy-relevant insights. Actively seeking roles in GIS, remote sensing, and environmental analysis locally and internationally.

TECHNICAL SKILLS

GIS & Remote Sensing: Google Earth Engine (GEE), QGIS, ArcMap, Land Use/Land Cover Mapping, NDVI Analysis, Change Detection, Image Classification, Time-Series Analysis, Satellite Imagery Processing

Field & Data: GPS Field Surveys, Coordinate Plotting, Microsoft Excel (Data Visualization, Cleaning, Statistical Analysis), Cartographic Map Production

Programming: JavaScript (Google Earth Engine API) | Currently learning: Python for geospatial analysis

Languages: English (Fluent), Twi (Native)

WORK EXPERIENCE

GIS & Remote Sensing Analyst | Forestry Services Division *Sunyani, Bono Region* | Oct 2024 – Aug 2025

National Service Posting

- Conducted land use/land cover mapping and change detection using Google Earth Engine with Landsat satellite imagery, producing actionable outputs for forest management decision-making.
- Performed spatial analysis and professional map production using QGIS and ArcMap to support land use planning and resource development across the Bono Region.
- Carried out GPS-based field surveys across forest sites, collecting coordinates, measuring land areas, and validating satellite-derived classifications with ground truth data.
- Managed and analyzed environmental datasets in Microsoft Excel, including data cleaning, visualization, and statistical reporting for forestry project documentation.
- Contributed to policy recommendations on sustainable forest management by translating spatial analysis findings into clear, non-technical reports for stakeholders.

GIS & Remote Sensing Intern | Forestry Services Division *Sunyani, Bono Region* | Feb 2024 – May 2024

- Created land use/land cover maps and time-series charts from Landsat imagery using Google Earth Engine.
- Conducted spatial analysis and map production in QGIS and ArcMap to support ongoing forestry projects.
- Used GPS devices to collect geographic coordinates and measure land areas during field surveys.
- Cleaned and organized environmental datasets in Microsoft Excel to support team reporting.

KEY PROJECTS

Land Cover Change Analysis — Berekum Municipal Area (2002–2022)

- Analyzed 20 years of land use/land cover change using Landsat 7 and Landsat 8 imagery processed in Google Earth Engine, with NDVI thresholding ($\mu \pm 2\sigma$) and cross-sensor calibration.
- Classified five land cover types across multiple time periods, quantifying forest loss and agricultural encroachment to inform Forestry Services Division management decisions.
- Produced a non-technical policy brief translating findings for government stakeholders and resource planners.

Spatial Distribution of Pollutant Gases in Ghana & Health Implications (2019–2024)

- National-scale study mapping NO₂, SO₂, and CO concentrations across all 16 regions of Ghana using Sentinel-5P satellite data processed in Google Earth Engine and QGIS.
- Integrated WHO air quality benchmarks and regional healthcare data in a mixed-methods framework, identifying pollution hotspots and seasonal trends over a five-year period.
- Final year research project supervised by Dr. Ebenezer F. Amankwaa, University of Ghana.

EDUCATION

BA, Geography & Resource Development with Political Science | University of Ghana *Legon, Accra* | 2021 – 2024

Second Class Upper Division

Relevant Coursework: Remote Sensing, Geographic Information Systems, Principles & Methods of Cartography, Resource Analysis, Research Methods, Strategies of Development, Public Administration

WASSCE, General Arts | Sumaman Senior High School *Suma-Ahenkro, Bono Region* | 2017 – 2020

Elective Subjects: Geography, History, Government, Economics

REFERENCES

Available upon request.